

CV

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Gyu Min Lee (이규민)

Postdoctoral Researcher

Affiliation: KRIBB (Korea Research Institute of Bioscience and Biotechnology), Center for Synthetic Biology

Location: Daejeon, South Korea

Principal Investigator: Hyewon Lee, Ph.D.

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BIOGRAPHY

Postdoctoral researcher at KRIBB Center for Synthetic Biology, working on C1 microbe engineering with AI coding agents in the loop.

RESEARCH

Methanotroph and methylotroph engineering for C1 chemistry. Adaptive laboratory evolution (ALE) of *Methylobacterium extorquens* under methanol stress (~800 generations, PhD work at UNIST).

METHODS

Whole-genome sequencing + RNA-seq + AlphaFold3 structural analysis to read mutations as a coherent stress-response network.

TOOLS

Custom dry-lab tooling on top of AI coding agents — primer design, NGS verification, lab notebook automation — so the wet-lab cycle never has to wait.



EDUCATION

2019.03 - 2026.02

Ph.D. (Integrated M.S./Ph.D.), Chemical Engineering

Ulsan National Institute of Science and Technology (UNIST)

Systems Biology and Machine Learning Lab, advisor: Donghyuk Kim, Ph.D.

Dissertation: Systems-level characterization of adaptive responses to methanol stress in *Methylobacterium extorquens*.

2011.03 - 2017.02

B.S., Genetic Engineering

Kyung Hee University

EXPERIENCE

2026.03 - present

Postdoctoral Researcher

KRIBB, Center for Synthetic Biology

C1 team led by Hyewon Lee, Ph.D.

Methanotroph strain engineering, multi-omics pipelines, agentic tooling for the C1 team.

2019.03 - 2026.02

Graduate Researcher

UNIST, Department of Chemical Engineering

Systems Biology and Machine Learning Lab (Donghyuk Kim)

Methylobacter ALE, pan-genome and DEG analysis, bacteriophage and plant-pathogen comparative genomics.

SKILLS

WET LAB

Adaptive laboratory evolution (ALE)

Methylobacter and methanotroph cultivation

Allelic exchange and mutation reconstruction

Growth phenotyping

RNA-seq library prep

OMICS AND MODELING

Whole-genome sequencing analysis

Pan-genome comparative analysis

RNA-seq and DEG analysis

AlphaFold3 structure-function interpretation

COBRApy genome-scale metabolic modeling

ENGINEERING

Python and R omics pipelines
Visualization automation
Claude Code agent workflows for synthetic biology
GitHub Actions CI and cron jobs
TypeScript / Astro for research tooling

SELECTED PUBLICATIONS

1. Systems-Level Dissection of Methanol Adaptation in *Methylobacterium extorquens* · Gyu Min Lee (2026, BITE (under revision)). link (<https://scholar.google.com/scholar?cluster=17800352162623955715&hl=en&oi=scholar>).
2. Characterization and therapeutic potential of newly isolated bacteriophages against *Staphylococcus* species in bovine mastitis · Jae-hyun Cho, Gyu Min Lee, Seyoung Ko, Youngju Kim et al. (2025, Journal of Virology). link (<https://journals.asm.org/doi/abs/10.1128/jvi.01901-24>).
3. Integrated genomic and transcriptomic Insights into methanol tolerance mechanisms in *Methylobacterium extorquens* AM1, identifying key targets for strain engineering · Gyu Min Lee, Khoi Nhat Pham, Ina Bang, Seyoung Ko et al. (2025, Journal of Biological Engineering). link (<https://link.springer.com/article/10.1186/s13036-025-00557-1>).
4. Structural and Transcriptomic Signatures of Adaptive Optimization in *Methylobacterium extorquens* PA1 $\Delta fdh234$ under Methanol Stress · Gyu Min LEE, KIM Jaehyung, KIM Donghyuk (2025, 한국생물공학회 학술대회). link (<https://www.dbpia.co.kr/Journal/articleDetail?nodeId=NODE12491336>).
5. Deciphering High-Methanol Adaptation in *Methylobacterium extorquens* AM1 through Genomic and Transcriptomic Analyses · Gyu Min LEE, PHAM Khoi, BANG Ina, KO Seyoung et al. (2025, 한국생물공학회 학술대회). link (<https://www.dbpia.co.kr/Journal/articleDetail?nodeId=NODE12491334>).
6. Pan-genome Analysis Revealed Comparable Genomic Characteristics of Central Metabolic Pathways in *Methylobacterium extorquens* · Gyu Min LEE, Zoe SCOTT-NEVROS, LEE Sang-Mok, KIM Donghyuk (2025, 한국생물공학회 학술대회). link (<https://www.dbpia.co.kr/Journal/articleDetail?nodeId=NODE12491335>).
7. Investigation of Evolutionary Strategies of *M. extorquens* AM1 for Enhanced Methanol Tolerance through Systems Biology Approaches · Gyu Min LEE, KO Seyoung, BANG Ina, KIM Donghyuk (2024, 한국생물공학회 학술대회). link (<https://www.dbpia.co.kr/Journal/articleDetail?nodeId=NODE11790945>).
8. Pan-genome Analysis Reveals Comparative Genomic Features of Central Metabolic Pathways in *Methylobacterium extorquens* · Gyu Min Lee, Zoe K Scott-Nevros, Sang-Mok Lee, Donghyuk Kim (2023, Biotechnology and Bioprocess Engineering). link (<https://link.springer.com/article/10.1007/s12257-022-0154-1>).

See full publication list →

AWARDS & SERVICE

To be added

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